Understanding Ammonium Salts Formation in CO_2 Transport with Impurities: A Gibbs Minimization Approach

Sviatoslav Eroshkin[1], Maria Eduarda Dias Serenario[1], Bruno Burmas[1], Sven Morten Hesjevik[1], and Anne Marie Koren Halvorsen[1]

[1]Equinor

Abstract

During experimental work involving CO_2 transport with impurities, the formation of ammonium salts was observed; however, the specific reaction pathways remain under investigation. This article presents a theoretical framework that aims to evaluate possible reaction pathways that could explain these phenomena using the Gibbs minimization approach. By analyzing the interactions between various impurities, this study provides insights into the conditions that favor ammonium salt formation, contributing to a deeper understanding of the chemical behavior in CO_2 transport systems.

1 Introduction

After carbon dioxide (CO_2) is captured from emission sources, it undergoes several steps to prepare for transport. Usually, CO_2 is captured using an amine unit, where it is separated from the gas stream. In the following step, it is essential to reduce moisture from CO_2, as the presence of liquid water can lead to severe complications during transport. The drying process typically involves compressing and cooling the gas to condense a portion of the water vapor. After condensation, the remaining moisture is eliminated through adsorption, with the most common adsorbents being molecular sieve and silica gel. The dehydration process is a standard procedure that has been effectively utilized in natural gas dehydration for many years.

In addition to moisture, the captured CO_2 may contain various impurities that need to be addressed to ensure it meets the quality standards required for safe transport and storage. The most prevalent impurities found in CO_2 include hydrogen sulfide (H_2S), sulfur dioxide (SO_2), nitrogen oxides (NO and NO_2), water (H_2O), oxygen (O_2), and amines. In total, a significant range of impurities can be present, each potentially contributing to corrosion. For example, the main impurities that could be present are outlined in the Northern Lights specification [3].

This underscores the necessity of robust modeling to predict the behavior of CO_2 and its impurities, as it is impractical to test every possible composition. While available equilibrium models may not provide precise practical answers, they offer valuable guidance on the potential states of the gas mixture and indicate whether certain chemicals are likely to form under specific conditions. By utilizing these models, one can streamline the testing process, focusing on the most relevant compositions and mechanisms.

2 Methods

2.1 Introduction to the Developed Model

The primary advantage of the Gibbs minimization model lies in its ability to perform calculations without specification of individual reactions. This feature is particularly advantageous for complex systems, such as CO_2 transport with impurities, where unknown reaction pathways may exist or where reaction mechanisms are not fully understood. This model operates under the principle that thermodynamics is indifferent to the specific pathway.

In the context of chemical equilibrium, the system's final stable state corresponds to the minimum of the total Gibbs free energy at a given temperature and pressure. Each potential chemical species that can form from these elements possesses a known standard Gibbs energy of formation G_0^f [kJ/mole], which is typically obtained from tabulated thermodynamic data, for example, ref. [4]. The total Gibbs free energy of a given mixture is then calculated as the sum of the partial molar Gibbs free energies (chemical potentials) of all constituent species, weighted by their respective molar amounts. The objective of the calculation is to determine the unique combination of species' molar amounts that satisfies the elemental mass balance constraints and yields the absolute lowest total Gibbs free energy for the entire system. If the optimization algorithm is robust, it will directly find the global minimum of the Gibbs energy, ensuring the true equilibrium state.

The model requires the inclusion of all species that could potentially arise from the chemical reactions. Therefore, it is crucial to identify which species can form and to understand which may be kinetically inhibited or unable to form due to non-equilibrium conditions. If any important species are omitted, the calculated equilibrium may be incomplete or inaccurate. Additionally, while the model can predict the equilibrium state that the system aims to achieve, it does not guarantee that the system will reach this state, as kinetics is not incorporated into the model.

The model was implemented using the open-source tool NeqSim, which is designed for simulating phase equilibria and thermodynamic properties in chemical processes. The implementation of the model can be accessed online, using AcidWatch tool, allowing users to use the models for CO_2 transport chemical reactions [1], [2].

2.2 Numerical Description

Gibbs free energy is a fundamental thermodynamic property that provides insight into thermodynamic feasibility of chemical reactions and phase transitions. The general formula for the total Gibbs free energy of a system is expressed as:

$$G_{\text{total}} = \sum_{i=1}^{N} n_i G_i \tag{1}$$

where

- G_{total}: Total Gibbs free energy of the system [kJ]
- N: Total number of components in the system [-]
- n_i: Number of moles of component i [mol]
- G_i: Gibbs free energy of component i [kJ/mol].

Eq. 1 sums the contributions of each component's Gibbs free energy, weighted by the number of moles present. It allows calculating the overall energy state of the system, which is crucial for determining its stability and the direction of spontaneous processes. To calculate Gibbs free energy for each component, one can use the following expression:

$$G_i = G^0_{f,298} + RT \ln\left(\frac{f_i}{P}\right) \tag{2}$$

where

- $G^0_{f,298}$: Standard Gibbs free energy of formation at 298 K [kJ/mol]
- R: Universal gas constant [kJ/(mol·K)]
- T: Temperature [K]
- f_i: Fugacity of component i [bar]
- P: Pressure [bar]

In Eq. 2, G_i^{mix} represents the energy associated with forming one mole of the component from its elemental state at standard conditions. The modified Gibbs free energy for each component can be expressed as:

$$G_i^{\text{mix}} = G^0_{f,i} + RT \ln(y_i) + RT \ln(\phi_i) + RT \ln(P) \tag{3}$$

where

- y_i: Component molar fraction [-]
- ϕ_i: Fugacity coefficient [-].

For the calculation of fugacity coefficient in Eq. 3, the Soave Redlich Kwong Equation of State (SRK EoS) was used using NeqSim [2].

2.3 Numerical Solution

In constrained optimization problems, such as determining the equilibrium composition of a chemical system where elemental balance must be satisfied, Lagrangian multipliers λ_e serve as mathematical tools that facilitate the incorporation of constraints directly into the objective function. This is achieved by constructing a new function, known as the Lagrangian, which combines the function to be minimized (in this case, the total Gibbs free energy) with the relevant constraint equations (the elemental mass balances). By setting the partial derivatives of this Lagrangian function with respect to the variables, specifically, the moles of species and the Lagrange multipliers, to zero, the constrained minimization problem is effectively transformed into an unconstrained system of equations. Thus, the important variables for the numerical solution of the system are the following:

1. **Objective function for component i:**

$$f_i = G^0_{f,298} + RT\ln(\phi_i) + RT\ln(y_i) + RT\ln(P) - \sum_{j=1}^{N_e} \lambda_e \nu_{ej} \tag{4}$$

   where

   - ν_{ej}: Number of moles of element e in molecule j.
   - λ_e: Lagrangian multiplier for element e.
   - N_e: Number of elements.

2. **Unknowns (x):** $y_1, y_2, \ldots, y_N, \lambda_1, \lambda_2, \ldots, \lambda_{N_e}$

3. **Minimization Vector $F(x)$:** $F = \{f_1, f_2, \ldots, f_{N_e}, \text{elemental mass balance}\}$

4. **Jacobian matrix J_x:** $J_x = \frac{\partial f_k}{\partial x_i}$

To determine the equilibrium composition of a chemical system, the Newton-Raphson iterative method is employed to find the roots of a system of nonlinear equations. This method initiates with an initial guess for the unknown variables and iteratively refines this composition until a satisfactory solution is achieved. Central to this process are the system's governing equations, which are organized into a minimization vector function $F(x)$.

At each step of the Newton-Raphson iteration, it is essential to not only know the current values of the functions $F(x)$, which indicate the distance from equilibrium, but also to understand how sensitive these functions are to changes in the variables involved. This sensitivity is quantified by the Jacobian matrix. The Jacobian is essentially a matrix containing all the first-order partial derivatives of each function f_k with respect to each variable

x (e.g., $n_1, n_2, n_3, \ldots, \lambda_1, \lambda_2, \ldots$). Below the key elements of the Jacobian matrix are shown.

The diagonal elements of the Jacobian matrix are given by:

$$J_{ii} = RT\left(\frac{1}{n_i} - \frac{1}{n_{\text{total}}}\right) + \frac{\partial \phi_i}{\partial n_i} \tag{5}$$

The off-diagonal elements are:

$$J_{ij} = -RT\left(\frac{1}{n_{\text{total}}}\right) + \frac{\partial \phi_i}{\partial n_j} \tag{6}$$

For elements, the derivatives with respect to the Lagrange multipliers are given by:

$$J_{iN+k} = -\nu_{ei} \tag{7}$$

Once the Jacobian is constructed, the minimization process can be carried out using the Newton-Raphson method. The core of the Newton-Raphson iteration involves solving a system of equations:

$$J(x^k)\Delta x^k = -f(x^k) \tag{8}$$

Once Δx^k is calculated, the composition is updated using the formula:

$$x^{k+1} = x^k + \alpha \Delta x^k \tag{9}$$

In this equation, α represents the step length, which is often set to 1 initially but may be adjusted as necessary to ensure stability. This iterative process continues, recalculating Δx^k until the changes are sufficiently small (e.g., below a predefined tolerance), indicating that a solution, the equilibrium composition, has been achieved.

3 Results and Discussion

3.1 Reactions with H_2S and Formation of Ammonium Salts

The combinations of sulphur dioxide (SO_2), nitrogen dioxide (NO_2), hydrogen sulphide (H_2S), oxygen (O_2), and water (H_2O) were observed to potentially facilitate the formation of ammonium salts. Utilizing the Gibbs minimization model enables the analysis of the specific conditions under which these ammonium salts may arise. The simulation results indicated that increased concentration of H_2S favours the formation of ammonium bisulphate (NH_4HSO_4).

For this analysis, the following impurity concentrations were considered at a pressure of 100 bar and a temperature of 25 °C:

- NO_2: 15 ppm
- SO_2: 0-5 ppm
- O_2: 30 ppm
- H_2O: 10 ppm

The concentration of H_2S was systematically varied in the model from 0 to 45 ppm to evaluate the resulting product distribution. The results, illustrated in Figure 1, demonstrate the impact of H_2S concentration on the formation of ammonium bisulphate (NH_4HSO_4).

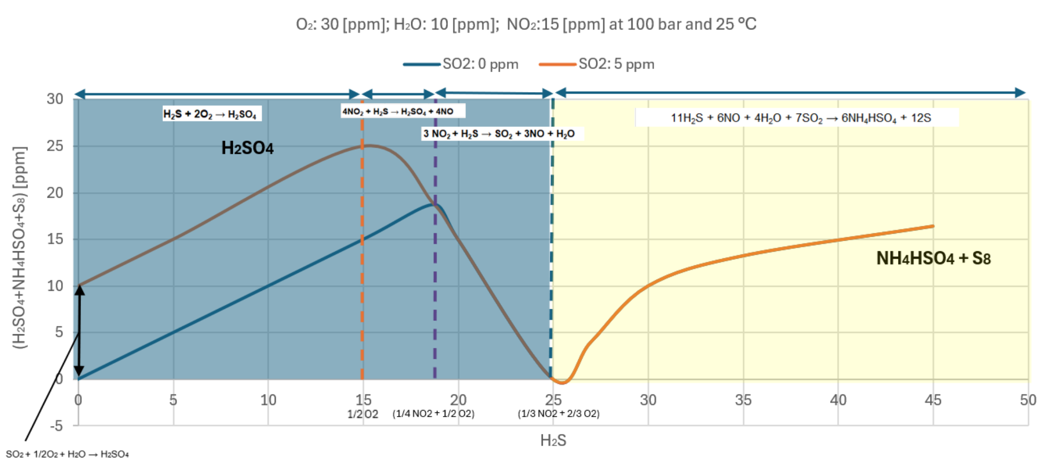

Figure 1: Gibbs minimization model prediction for the following composition: NO_2: 15 [ppm], SO_2: 0-5 [ppm], O_2: 30 [ppm], H_2O: 10 [ppm] at 100 bar and 25 °C case with varying H_2S concentration.

The results depicted in Figure 1 can be explained through the following expected reactions:

$$3\mathrm{NO_2} + \mathrm{H_2S} \rightarrow \mathrm{SO_2} + 3\mathrm{NO} + \mathrm{H_2O} \tag{10}$$

$$\mathrm{NO_2} + \mathrm{SO_2} + \mathrm{H_2O} \rightarrow \mathrm{H_2SO_4} + \mathrm{NO} \tag{11}$$

$$\mathrm{NO} + \frac{1}{2}\mathrm{O_2} \rightarrow \mathrm{NO_2} \tag{12}$$

Combination of Eq. 11 and Eq. 12, yields:

$$4\mathrm{NO_2} + \mathrm{H_2S} \rightarrow \mathrm{H_2SO_4} + 4\mathrm{NO} \tag{13}$$

Combination of Eq. 13 with Eq. 12 yields:

$$\mathrm{H_2S} + 2\mathrm{O_2} \rightarrow \mathrm{H_2SO_4} \tag{14}$$

Eq. 14 indicates that if both O_2 and NO_2 are present, H_2S will react with O_2 to form H_2SO_4. NO_2 will act in intermediate steps, meaning it will not be consumed but is necessary for the reaction to proceed. From Eq. 14, one can conclude the following: if H_2S [ppm] $\leq \frac{1}{2}$ O_2 [ppm] and NO_2 is present, Eq. 14 will dominate the reaction pathway.

From Eq. 13 and Eq. 14, one can further deduce: if $\frac{1}{2}$ O_2 [ppm] $\leq$ H_2S [ppm] $\leq \frac{1}{4}$ NO_2 [ppm], Eq. 13 will be the prevailing reaction.

By integrating the derived conditions, one can get the condition for maximum H_2SO_4 formation, expressed as: H_2S [ppm] $= \frac{1}{4}$ NO_2 [ppm] $+\frac{1}{2}$ O_2 [ppm].

If one further combines Eq. 10 and Eq. 12:

$$\frac{3}{2}\mathrm{O_2} + \mathrm{H_2S} \rightarrow \mathrm{SO_2} + \mathrm{H_2O} \tag{15}$$

Considering Eq. 10 and Eq. 15: if $\frac{1}{2}\,O_2$ [ppm]$+\frac{1}{4}\,NO_2$ [ppm] $\leq H_2S$ [ppm] $\leq \frac{2}{3}\,O_2$ [ppm] $+ \frac{1}{3}\,NO_2$ [ppm], then reaction 10 will proceed. This is due to the highly reactive nature of H_2S, which, when present in excess, will utilize the maximum amount of NO_2 for oxidation. In Eq. 10, only 3 moles of NO_2 react with 1 mole of H_2S, while in Eq. 13, 4 moles of NO_2 are required for the oxidation of H_2S. When H_2S [ppm] $= \frac{2}{3}\,O_2$ [ppm] $+ \frac{1}{3}\,NO_2$ [ppm], minimum

H_2SO_4 should be produced (under given pressure and temperature), and all H_2S will be converted to SO_2 according to Eq. 10.

When $H_2S > \frac{2}{3}O_2 + \frac{1}{3}NO_2$, H_2S will further react with products from Eq. 10:

$$5H_2S + 6\text{NO} + 4H_2\text{O} \rightarrow 6\text{NH}_3 + 5\text{SO}_2 \tag{16}$$

$$\text{NH}_3 + \text{H}_2\text{S} \rightarrow \text{NH}_4\text{SH} \tag{17}$$

$$\text{NH}_4\text{SH} + 2\text{SO}_2 \rightarrow \text{NH}_4\text{HSO}_4 + 2\text{S} \tag{18}$$

The reactions outlined above will yield the following overall reaction Eq. 19:

$$11H_2S + 6\text{NO} + 4H_2\text{O} + 7\text{SO}_2 \rightarrow 6\text{NH}_4\text{HSO}_4 + 12\text{S} \tag{19}$$

The presented theory is valid for mild temperatures. According to the Gibbs minimization model, Eq. 19. becomes more favorable with lower temperature. Thus, under lower temperatures, H_2SO_4, NH_4HSO_4, and elemental sulfur (S, S_8) can be produced at the same time, for example as shown in Figure 2. The summary of the developed predictions are represented in Table 1.

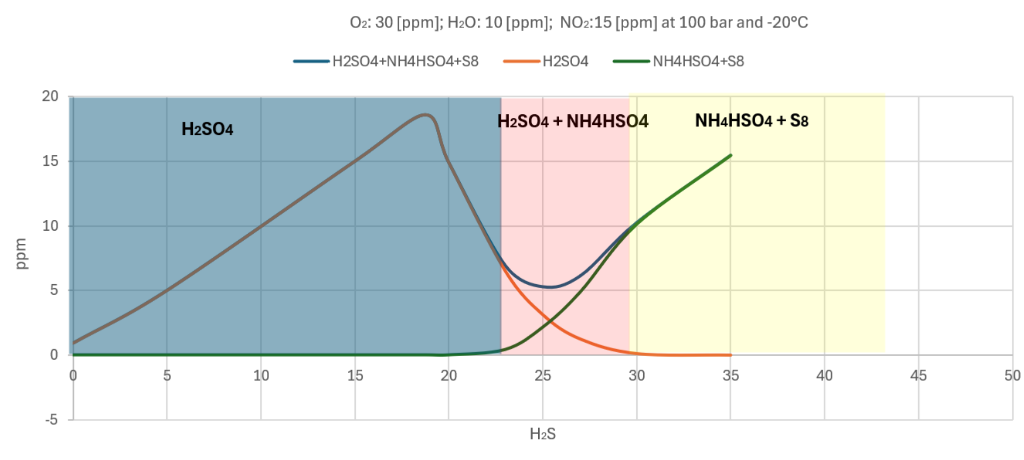

Figure 2: Gibbs minimization model prediction for NO_2: 15 ppm, SO_2: 0-5 ppm, O_2: 30 ppm, H_2O: 10 ppm at 100 bara and -20 °C case with varying H_2S.

Table 1: Summary of the reaction mechanisms according to Gibbs minimization model. Assuming NO_2 and SO_2 are present if not specified otherwise

H_2S [ppm]	Pipeline transport conditions (higher temperature)	Ship transport conditions (lower temperature)
0	NO_2 may react with SO_2 under specific conditions, and H_2SO_4 and HNO_3 may potentially be formed.	
$\leq \frac{1}{2}$ O_2 [ppm]	Production of H_2SO_4 from Eq. 14.	
$\leq \frac{1}{2}$ O_2 [ppm] + $\frac{1}{4}$ NO_2 [ppm]	- Production and maximization of H_2SO_4 from Eq. 13. -If SO_2 is present, the maximum of H_2SO_4 is at the point H_2S [ppm] = $\frac{1}{4}$ NO_2 [ppm] + $\frac{1}{2}$ O_2. -If SO_2 is not present, the maximum of H_2SO_4 is at the point H_2S [ppm] < $\frac{1}{4}$ NO_2 [ppm] + $\frac{1}{2}$ O_2 [ppm]	
$< \frac{2}{3}$ O_3 [ppm] + $\frac{1}{3}$ NO_2 [ppm]	Minimization of H_2SO_4. SO_2, NO, and H_2O are produced according to Eq.10 .	Minimization of H_2SO_4¨. Possible formation of ammonium salts
$= \frac{2}{3}$ O_3 [ppm] + $\frac{1}{3}$ NO_2 [ppm]	H_2SO_4 is at the minimum, mostly SO_2, NO and H_2O.	Formation of ammonium salts + H_2SO_4.
$> \frac{2}{3}$ O_3 [ppm] + $\frac{1}{3}$ NO_2 [ppm]	H_2SO_4 is at the minimum. Production of NH_4HSO_4 + S_8 from Eq. 19. Possible formation of other products with increase in temperature.	Gradual reduction of H_2SO_4 to minimum, formation of NH_4HSO_4 + S_8 from Eq. 19. Possible formation of other products with increase in temperature.

Comparison with Experiments

Recent research by Svenningsen, Morland, and Solberg [5] provided the tests that were conducted using a mixture that adhered to the Northern Lights CO_2 specification (2019): 30 ppm-mol H_2O, 10 ppm-mol SO_2, 10 ppm-mol NO_2, 10 ppm-mol O_2, and 9 ppm-mol H_2S. Under pipeline conditions at +4°C and 100 bar, the presence of NO_2 was found to trigger acid formation and precipitation of acids (H_2SO_4, HNO_3, and potentially S_8), resulting in a calculated corrosion rate of 2.8 µm/year, with no localized corrosion observed.

However, when the same test was repeated, with increased H_2S concentration, 60 ppm-mol, several reactions produced a white powder identified as a mixture of sulphur and ammonium salts.

The authors state that "the presence of ammonia (NH_3/NH_4^+) was unexpected, as the reactions leading to ammonia formation have not been previously observed in carbon capture and storage (CCS)-related experiments. In this context, the only nitrogen source is nitrogen dioxide (NO_2) and its reaction products, such as nitric oxide (NO) and nitric acid (HNO_3)." The reaction pathway for ammonia formation was unknown and warranted further investigation in future studies. The developed theory was used to explain the observed behaviour (summary given in Table reftab:developedtheory).

The developed theory effectively represents the experimental findings; however, some deviations from the observed results were noted. While the formation of ammonium aligns well with theoretical predictions, it is crucial to approach all equilibrium model results with caution, as kinetic factors are not accounted for in the system. Rather than relying on precise numerical values, the focus should be on describing the theoretical behaviour of the system based on the modelling approach. This has been accomplished in the current work, but further research is necessary to validate the theory and its applicability.

Table 2: Developed theory comparison with experimental data

Impurities	**Results**	**Explanation according to the developed theory**
30 ppm-mol H_2O + 10 ppm-mol SO_2 + 10 ppm-mol NO_2 + 10 ppm-mol O_2 + 8 ppm-mol H_2S	- Presence of NO_2 triggers the acid formation reaction and precipitation of acids. - Consumption of H_2S when NO_2 is injected. - Indication of acid formation: both H_2SO_4 and HNO_3 were formed. - Calculated corrosion rate was 2.8 µm/year with no localized corrosion.	According to the developed theoretical framework for moderate temperatures, the maximum concentration of H_2S is defined as $\frac{2}{3}$ O_3 [ppm] + $\frac{1}{3}$ NO_2 [ppm] = 10 [ppm]. This indicates that as long as H_2S concentration is lower than approximately 10 ppm, there is a likelihood of H_2SO_4 formation through Eq. 13 and Eq. 14.
30 ppm-mol H_2O + 10 ppm-mol SO_2 + 10 ppm-mol NO_2 + 10 ppm-mol O_2 + 60 ppm-mol H_2S	- No chemical reaction when only H_2O and NO_2 are present. - Several chemical reactions with white powder with yellow tint formation when all impurities are tested. - EDS and XRD indicated a mixture of S_8 and ammonium salts: $(NH_4)_2(SO_4)$ and $(NH_4)HSO_4$. - Very low corrosion rate $\sim$0.5 µm/year with no localized corrosion.	The maximum concentration of H_2S is also calculated as 10 [ppm]. This indicates that once H_2S concentration exceeds approximately 10 ppm, there is a likelihood of ammonium salt formation through hydrosulphide formation according to the general path 19.

4 Conclusion

The Gibbs minimization model has been utilized for many years for studying reaction equilibria and is recognized for its ability to provide reliable predictions of chemical reactions in specific applications. The model's primary advantage lies in its ability to analyze complex systems, such as CO_2 with impurities, without the need to specify individual reactions. By focusing only on the thermodynamic principles that govern the system's behavior, it identifies the stable state corresponding to the minimum total Gibbs free energy at given temperatures and pressures. However, the model also has limitations. It requires the input of possible species that could form from the elements present; missing any potential species may lead to partial or incorrect equilibrium calculations. Additionally, the model does not account for kinetics, which is crucial for understanding the speed of reactions and the energy barriers that must be overcome for reactions to occur. As a result, while the model can predict the equilibrium state, it cannot guarantee that the system will reach that state.

Given the inherent limitations of the Gibbs minimization model, one should not rely solely on numerical outputs. Instead, the model can be employed as a tool to investigate the underlying reaction mechanisms that may arise during the transport of CO_2 in the presence of impurities. This approach allows for a more comprehensive understanding of the complex interactions and potential pathways that could influence the behavior of CO_2 under varying transport conditions.

One potential mechanism warranting investigation is the formation of ammonia and ammonium salts. Observations indicate that combinations of sulfur dioxide (SO_2), nitrogen dioxide (NO_2), hydrogen sulfide (H_2S), oxygen (O_2), and water (H_2O) during CO_2 transport can facilitate the formation of the ammonium salts. However, the specific reaction pathways involved remain largely unknown and require further investigation. By utilizing the Gibbs minimization model, one can analyze the mechanisms that lead to ammonium salts with the aforementioned components. The simulations indicated that a higher concentration of H_2S favors the formation of ammonium bisulfate (NH_4HSO_4). Notably, there exists a threshold concentration of H_2S that must be exceeded for ammonia and ammonium salts to form. This threshold is defined as $\frac{2}{3}$ O_2 (ppm) + $\frac{1}{3}$ NO_2 (ppm) at moderate temperatures; however, it decreases as the temperature is lowered.

According to the developed theoretical framework, ammonia is created through the reaction of NO with H_2S. In this process, nitrogen transitions from NO_2 (with a +4 state) to NO (with a +2 state), and subsequently, if sufficient H_2S is present, it further reduces to ammonia (NH_3 with a -3 state). Reaction pathways based on the Gibbs minimization model were proposed.

While ammonia can react with sulfuric or nitric acid, the developed model suggests an additional pathway: ammonia further reacts with H_2S to form ammonium hydrosulfide (NH_4SH), which can then react with SO_2 to produce NH_4HSO_4 and elemental sulfur. Several experiments were chosen to validate this theoretical proposal.

Although the model effectively represents the experimental findings, some deviations from observed results were noted, and more experimental work is needed to further validate the theoretical study. While the formation of ammonium salts aligns well with current theoretical predictions, it is essential to approach equilibrium model numerical results with caution due to the absence of kinetic considerations and uncertainty in the model parameters.

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